

CS & IT ENGINEERING

COMPILER DESIGN

PARSING

Lecture No.-03



By-venkatsir

Recap of Previous Lecture



Topic

?? LR(0) Parsing Table

LR(0) items

LR(0) items DFA → closure()
→ goto()



Topics to be Covered



Topic

??

$\Rightarrow$ LR(0) Parsing Table

Topic

??

$\Rightarrow$ LR(0) grammars

Topic

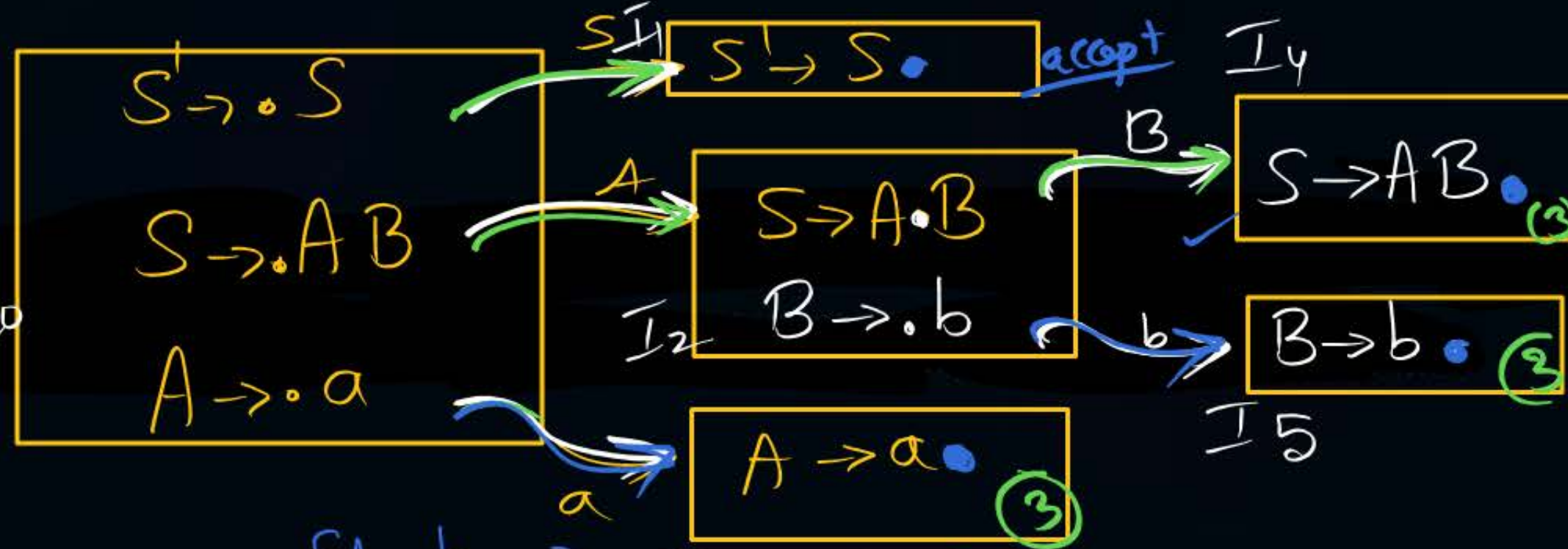
??

$\Rightarrow$ not LR(0) grammar

①

$S' \rightarrow S$

1 $S \rightarrow AB$
2 $A \rightarrow a$
3 $B \rightarrow b$



I_1, I_3, I_4, I_5

{Action} I_3

{GOTO}

	a	b	\$	S	A	B
I_0	$S_3 \checkmark$			1 $\checkmark$	2 $\checkmark$	
I_1			Accept $\checkmark$			
I_2		S_5 / r				4 $\checkmark$
I_3	r_2	r_2	r_2 / r_1			
I_4	r_1	r_1	r_1			
I_5	r_3	r_3	r_3			

$\checkmark$ $GOTO = (N.T + transitions)$

$\checkmark$ Action Table = terminal + transition (shift)

LR(0) Parsing Table

Reduce Action = $A \rightarrow XYZ$
 $3+3+3=9$

$S' \rightarrow S$

$A \rightarrow a$

$S \rightarrow AB$

$B \rightarrow b$

$r_2 \rightarrow$

r_1

r_3

Shift/Reduce Conflict

Reduce/Reduce Conflict

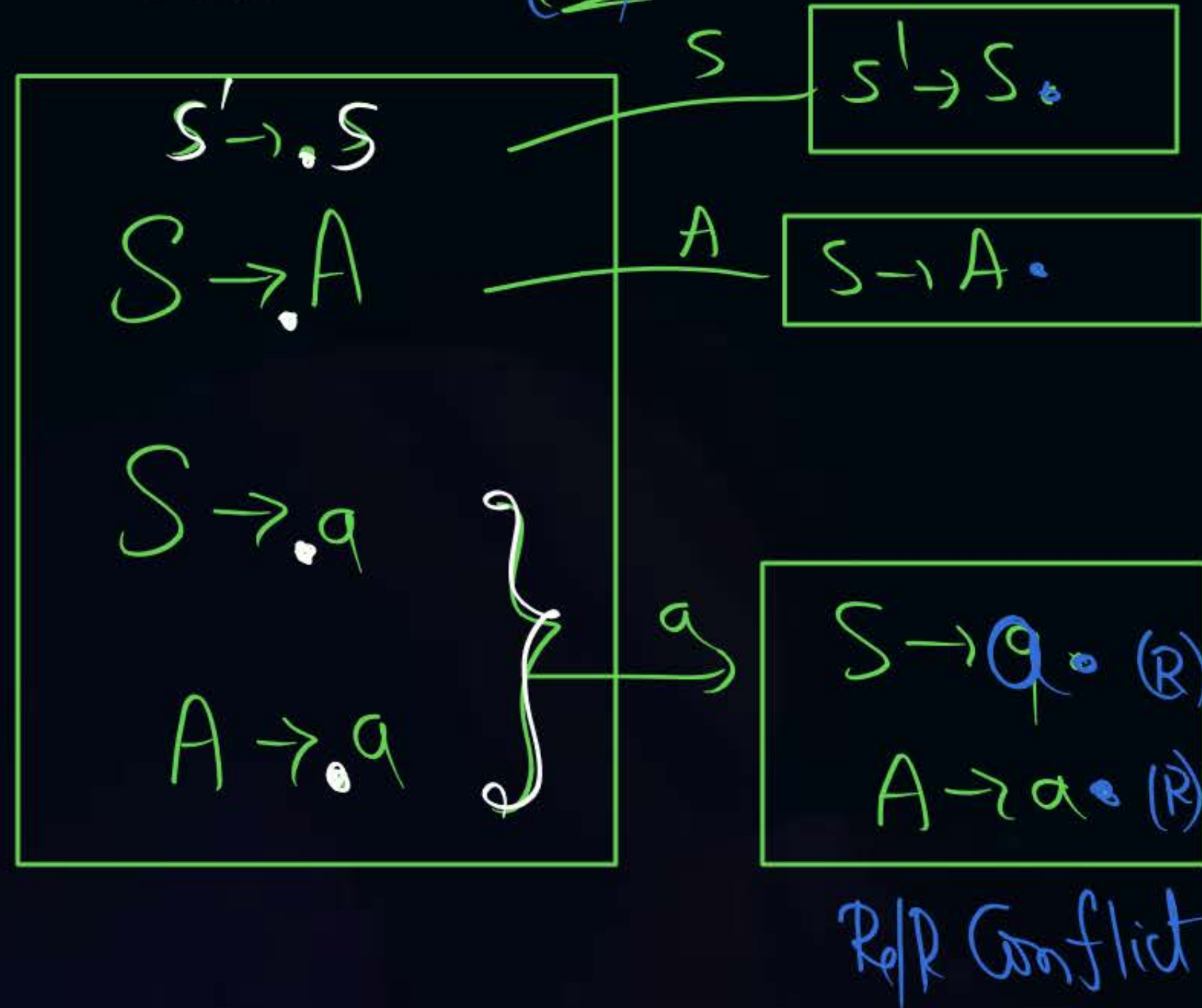
not LR(0) Grammar

LR(0) Grammar

Grammar Not having any conflict

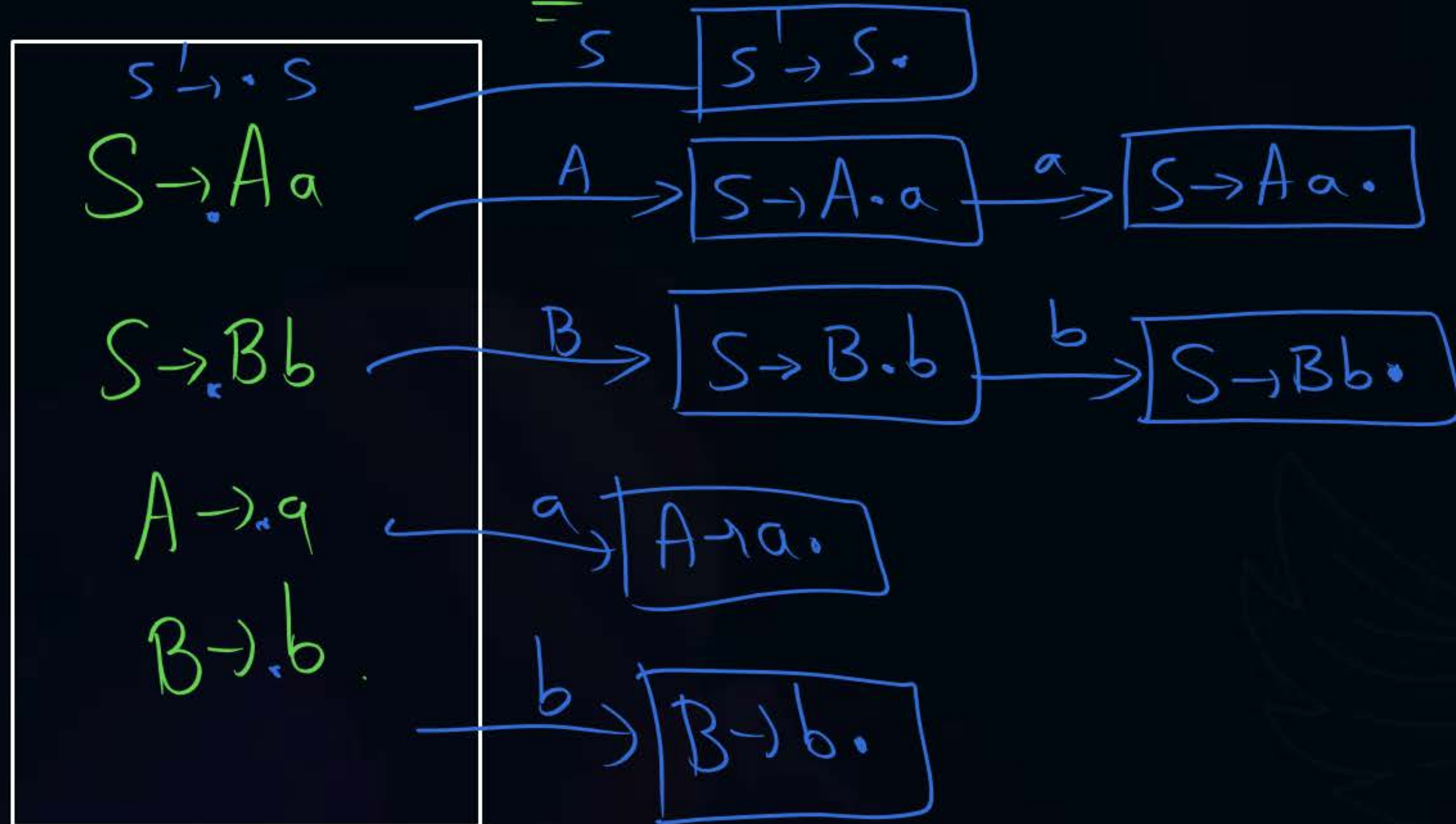
Shift Actions = terminal transitions in LR(0) or DFA
(2 shift actions)

#Q. LR(0) grammar (or) not?



not LR(0) grammar

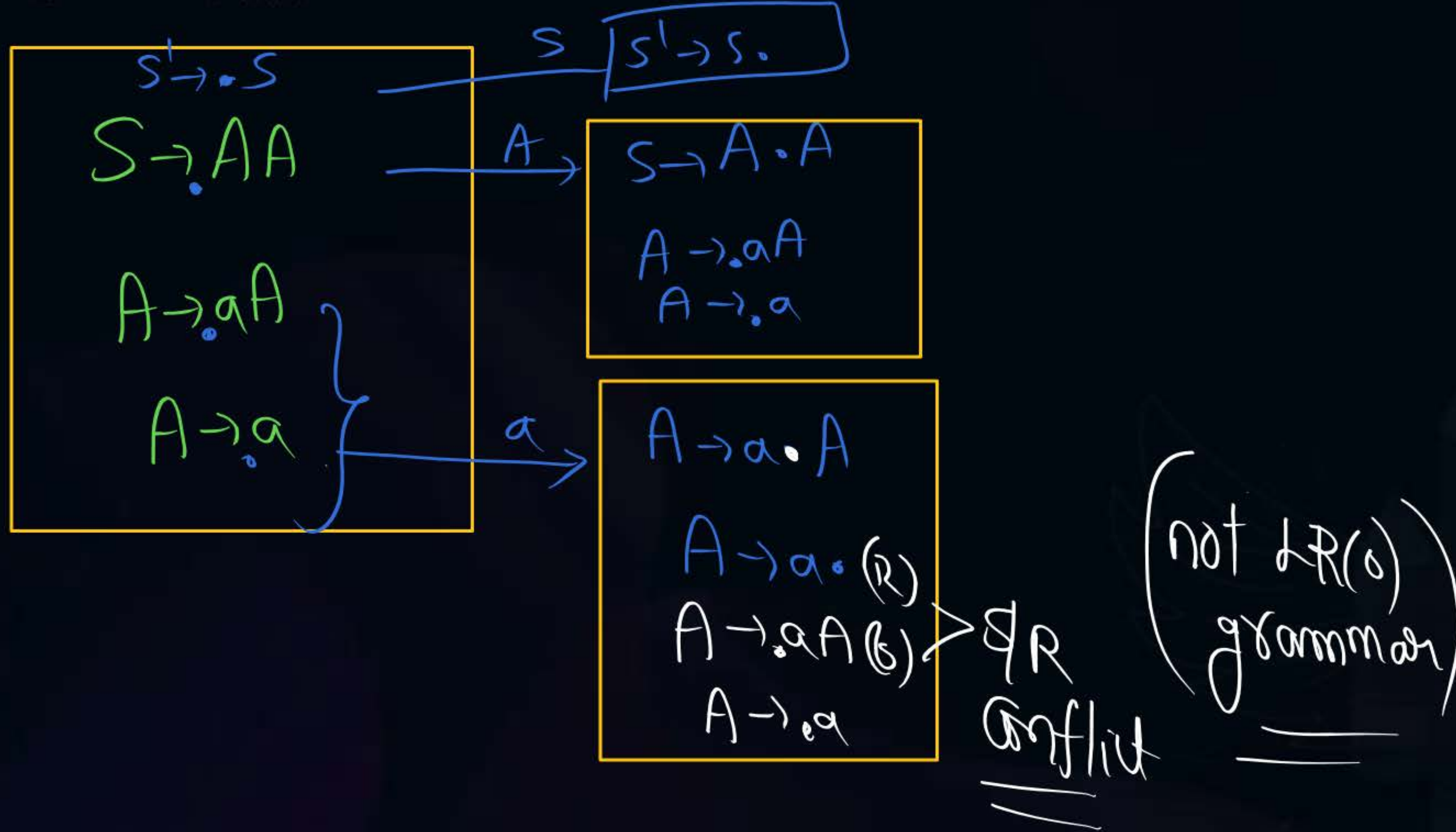
#Q. LR(0) grammar (or) not?.



No Conflict

LR(0) Grammar

#Q. LR(0) grammar or not?



NOTE

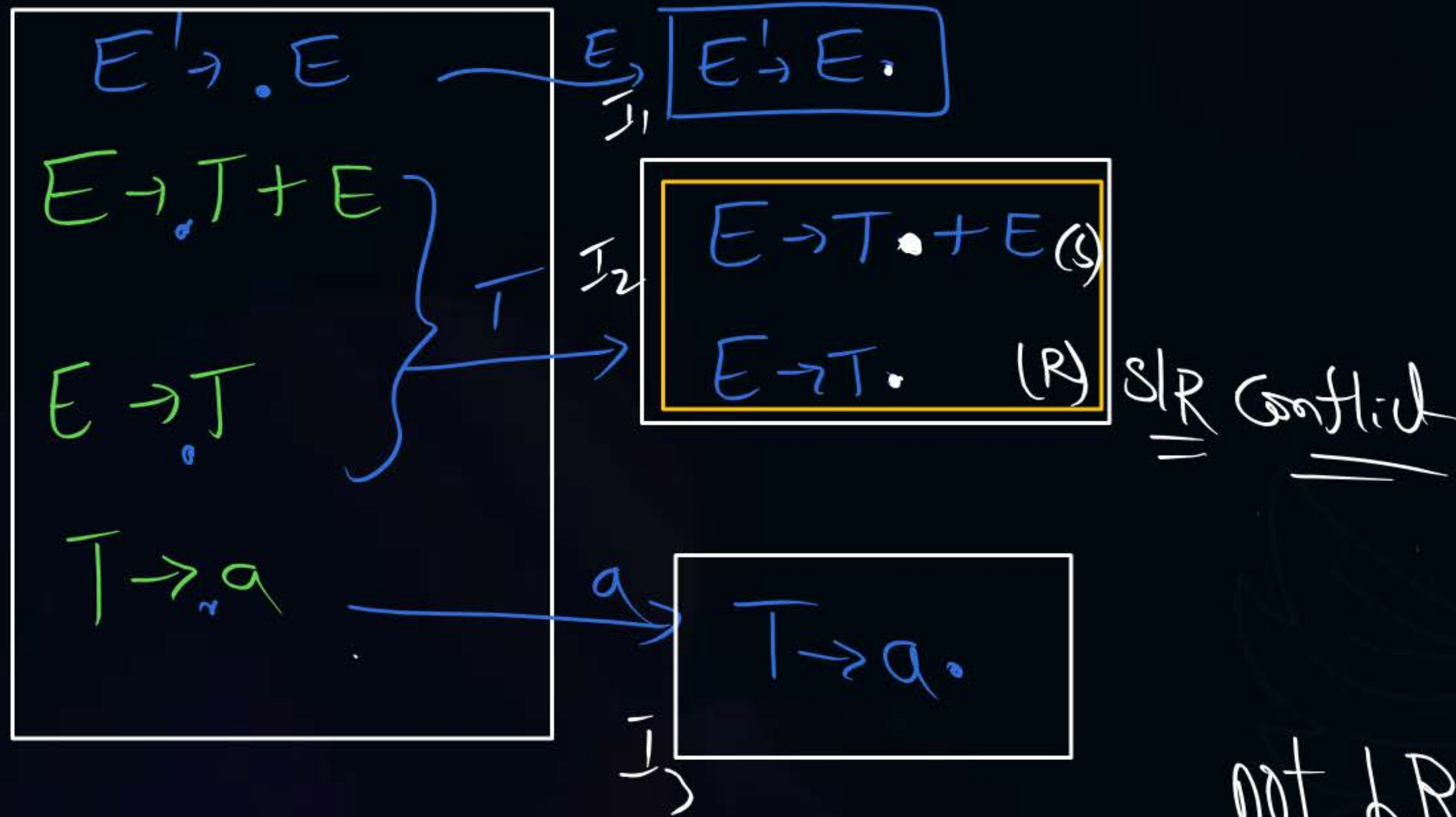
$s' \rightarrow \cdot s$
 $S \rightarrow \cdot a S a$
 $S \rightarrow \cdot b S b$ (Shift)
 $S \rightarrow \cdot \epsilon \cdot$ (Redu)

S/R Conflic

Not LR(0) grammar

$A \rightarrow \cdot \epsilon$
 $\Downarrow$
 $A \rightarrow \epsilon \cdot$

#Q. LR(0) grammar or not?.



not LR(0) grammar



Topic : Conflict

states

- S/R Conflict:

I_n

$$\begin{array}{l} A \rightarrow \alpha \cdot \overset{\text{terminal}}{a} B \text{ (Shift)} \\ B \rightarrow \gamma \cdot \text{ (Reduce)} \end{array}$$

- R/R Conflict:

$$\begin{array}{l} A \rightarrow \alpha \cdot \text{ (Reduce)} \\ B \rightarrow \gamma \cdot \text{ (Reduce)} \end{array}$$

I_m

(not LR(0)
grammar)

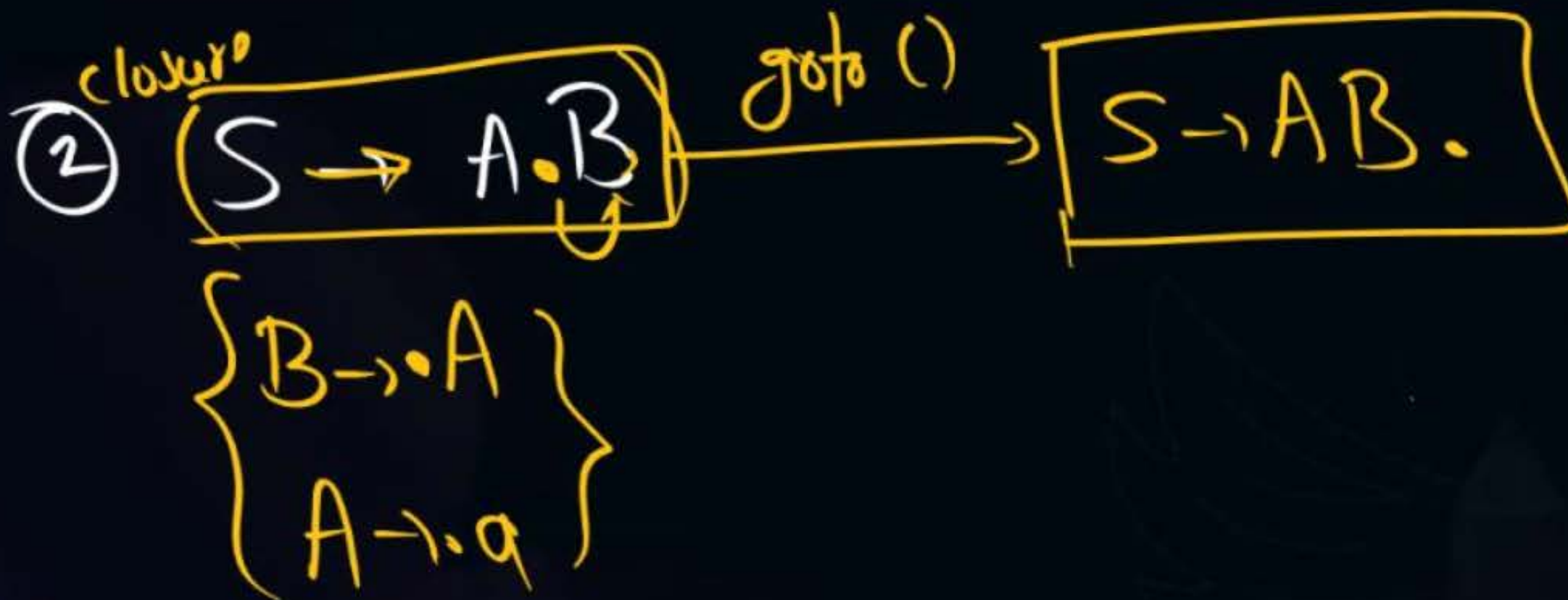


Topic : LR(0) Items

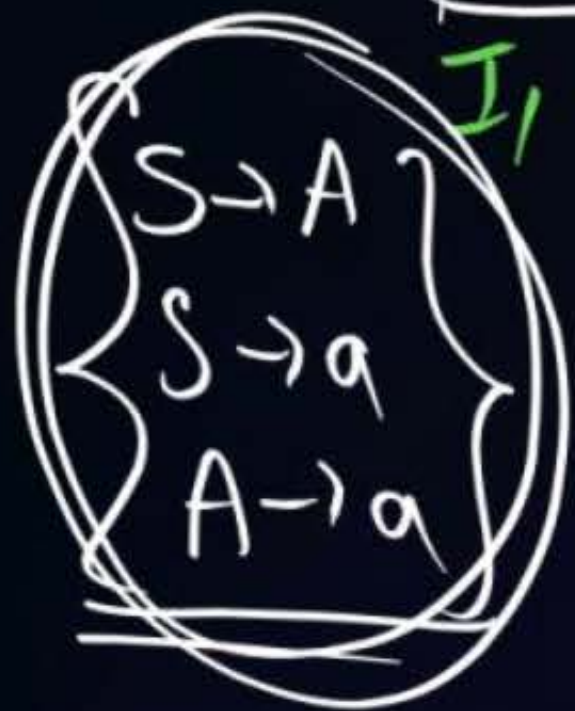
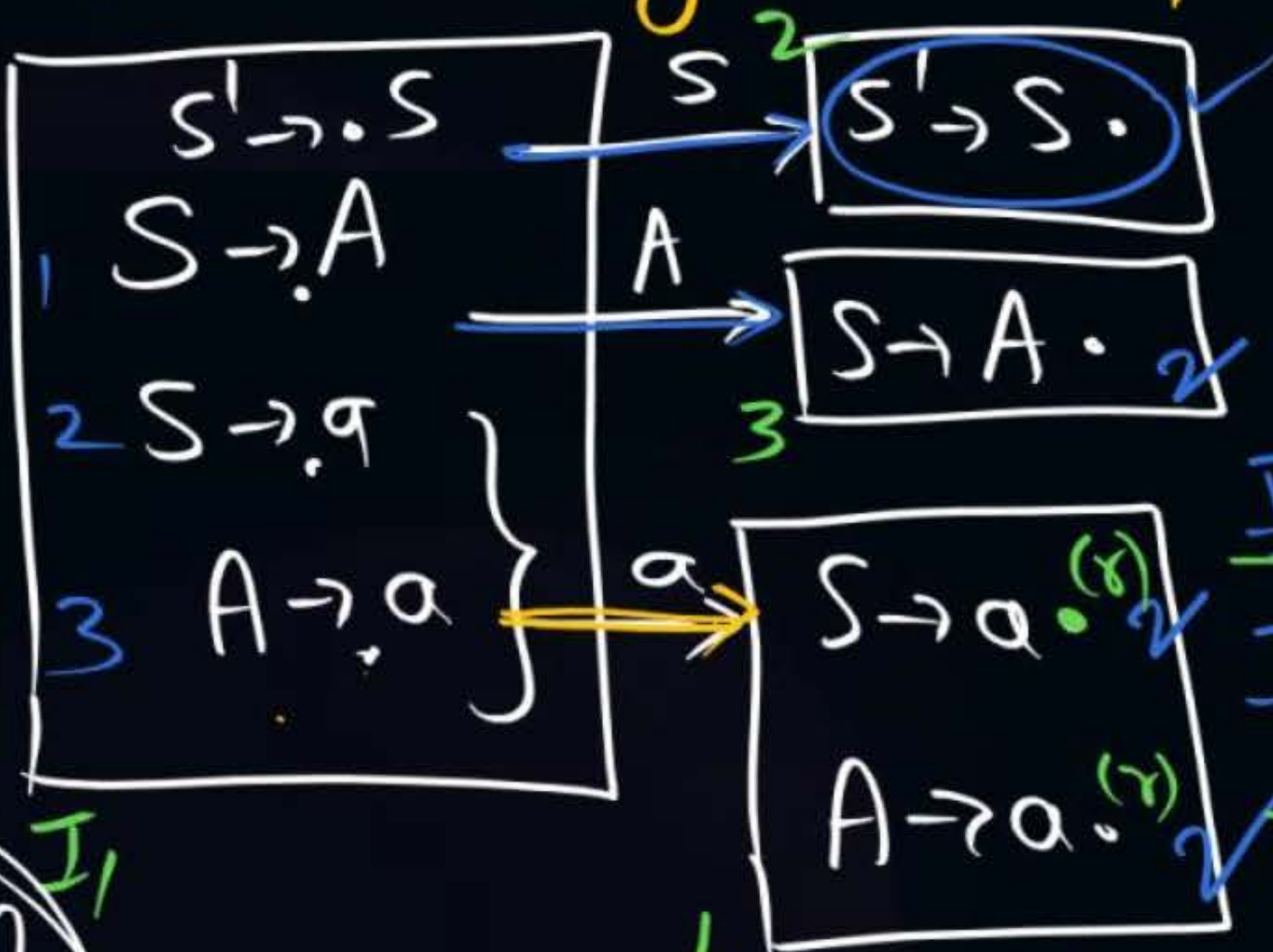
$$\textcircled{1} A \rightarrow x \cdot y z$$

$$B \rightarrow A$$

$$A \rightarrow a$$



(Q) How many states present in LR(0) Parsing table for following grammar



R/R Conflict to Reduce

LR(0) items DFA

not LR(0) grammar

1 shift action

2 GOTO entries

① Reduce/Reduce Conflict

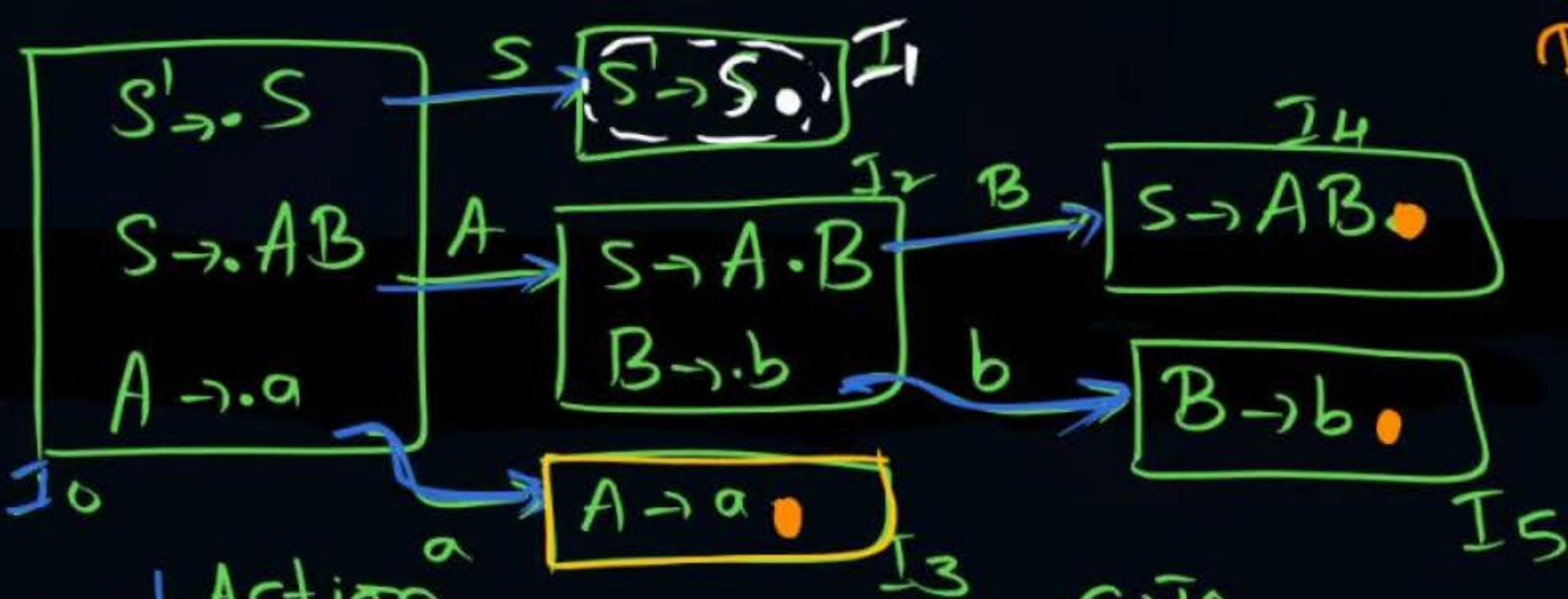
② Shift/Reduce Conflict

{LR(0) items
DFA}

- ①
- $S' \rightarrow S$
 - ① $S \rightarrow AB$
 - ② $A \rightarrow a$
 - ③ $B \rightarrow b$

$S \rightarrow AB$
 $A \rightarrow a$
 $B \rightarrow b$

States of DFA $\Rightarrow$



Reduce
 $\{I_3, I_4, I_5\}$

	Action			GoTo		
	a	b	\$	S	A	B
I ₀	S ₃			1 ✓	2 ✓	
I ₁			Accept ✓			
I ₂		S ₅ /r				4 ✓
I ₃	r ₂	r ₂	r ₂			
I ₄	r ₁	r ₁	r ₁			
I ₅	r ₃	r ₃	r ₃			

LR(0) Grammar

{LR(0) Parsing Table}

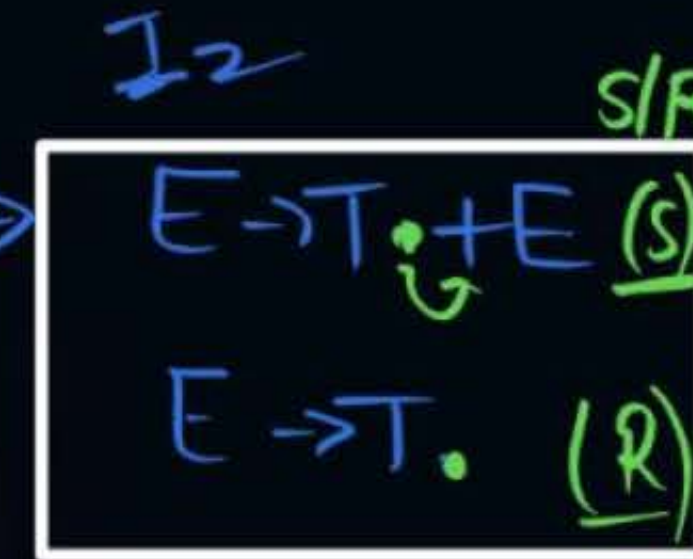
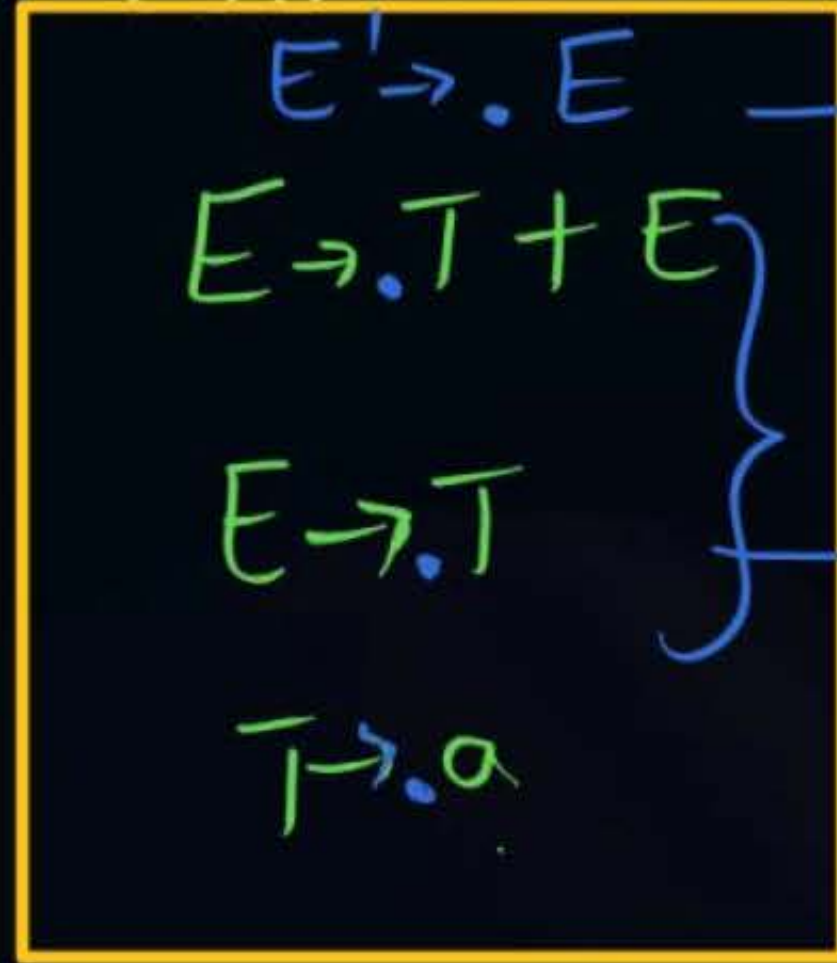
(no. of states) * (T + N + T + \$)

$$6 * (3 + 2 + 1)$$

$$6 * 6 = 36$$

#Q.

LR(0) grammar or not?



S/R Conflict

not LR(0) grammar

#Q.

LR(0) grammar or not?

$S' \rightarrow \cdot S$
 $S \rightarrow \cdot A A$
 $A \rightarrow \cdot a A$
 $A \rightarrow \cdot a$

Yes

No

I_1

$S' \rightarrow S \cdot$

I_2

$S \rightarrow A \cdot A$
 $A \rightarrow \cdot a A$
 $A \rightarrow \cdot a$

I_3

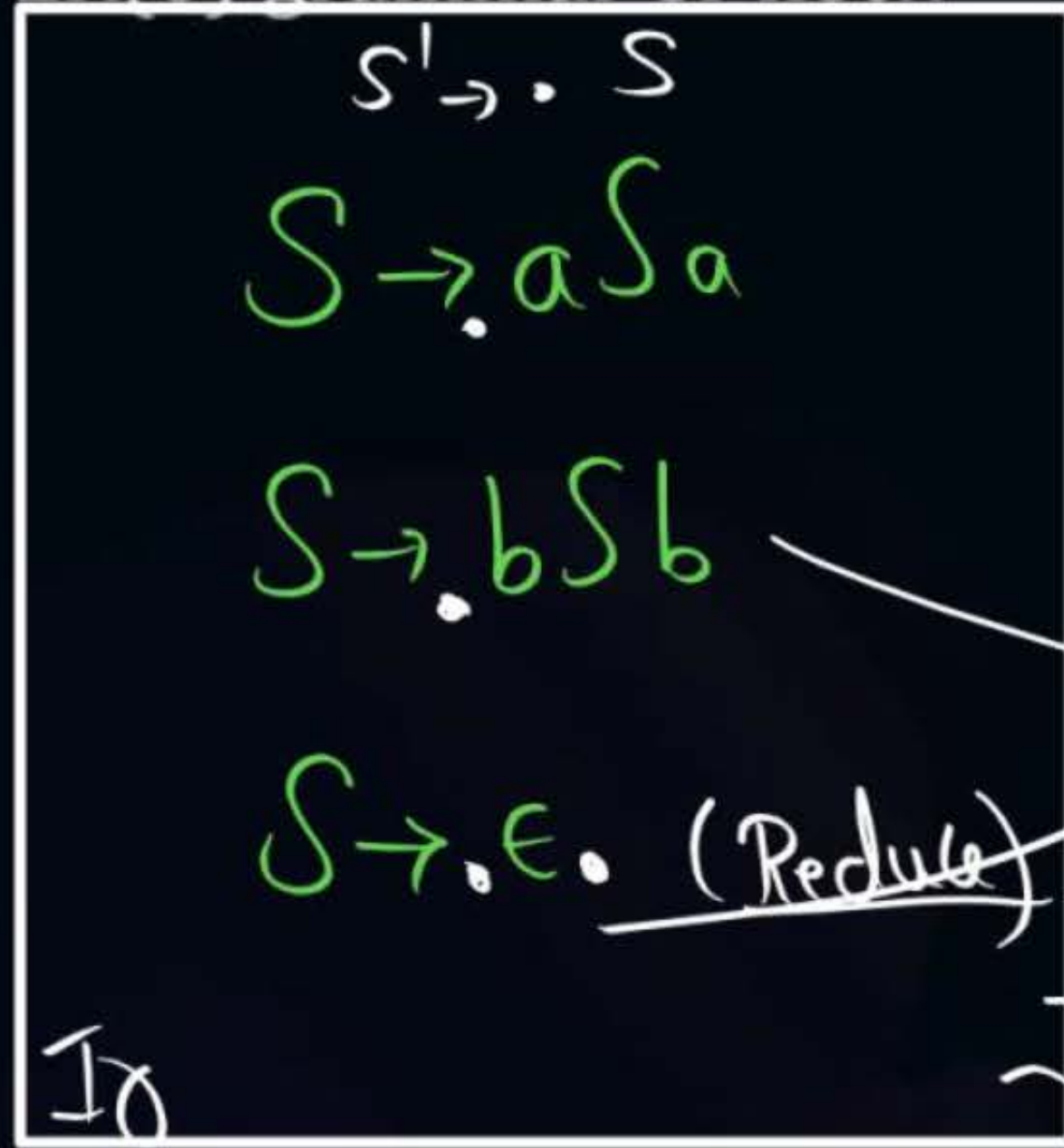
$\{ A \rightarrow a \cdot A \}$
 $\{ A \rightarrow a \cdot \}$
 $A \rightarrow \cdot a A (S)$
 $A \rightarrow \cdot a (S)$

SLR Conflict

not LR(0) Grammar

Yes
No

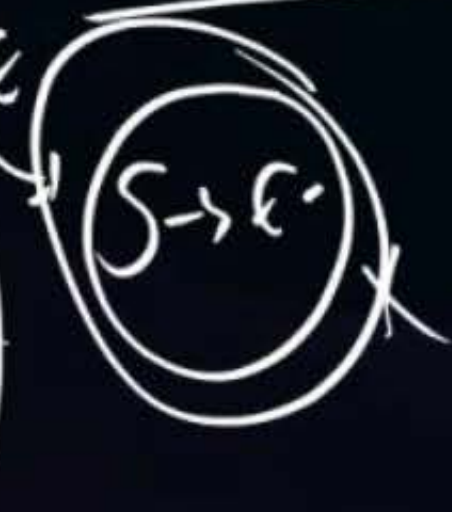
#Q. LR(0) grammar or not?



S/R Conflict

not LR(0) grammar

LR(0) item DFA



$$\begin{aligned} ① & \left\{ A \rightarrow \cdot \underline{\epsilon} \right\} \\ ② & \left\{ A \rightarrow \epsilon \cdot \right\} \end{aligned} \quad \underline{\underline{\text{Same}}}$$

#Q. LR(0) grammar or not?

$S' \rightarrow \cdot S$

$S \rightarrow \cdot a S b S$

$S \rightarrow \cdot b S a S (S)$

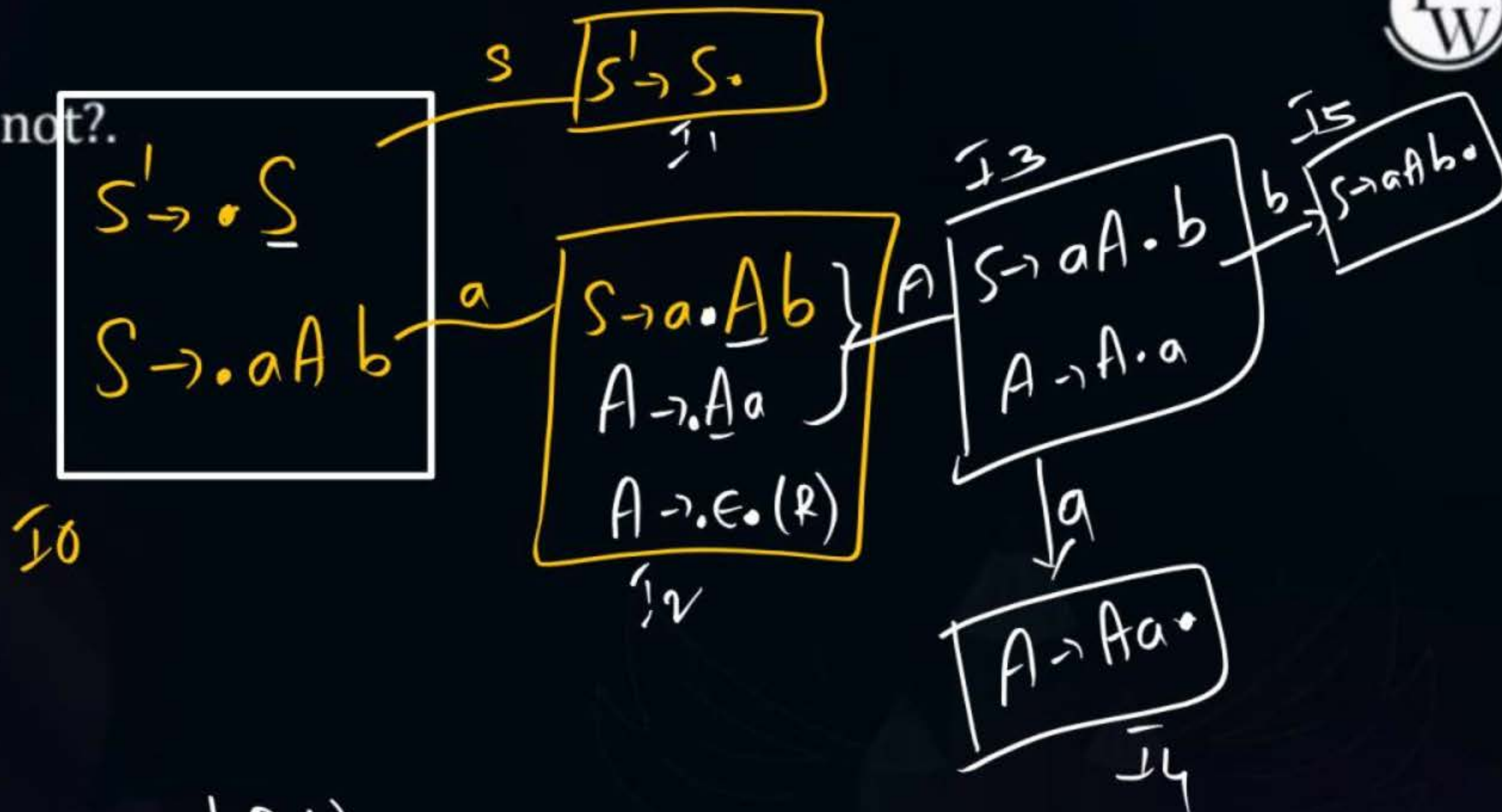
$S \rightarrow \cdot \epsilon \cdot (R)$

S/R Conflict

not LR(0) grammar

#Q. LR(0) grammar or not?

$$\left\{ \begin{array}{l} S \rightarrow aAb \\ A \rightarrow Aa \\ A \rightarrow \epsilon \end{array} \right\}$$



LR(0) grammar

#Q. LR(0) grammar or not?

$S' \rightarrow \cdot S$

$S \rightarrow \cdot A a A b$

$S \rightarrow \cdot B b B a$

$A \rightarrow \cdot \epsilon \cdot (R)$

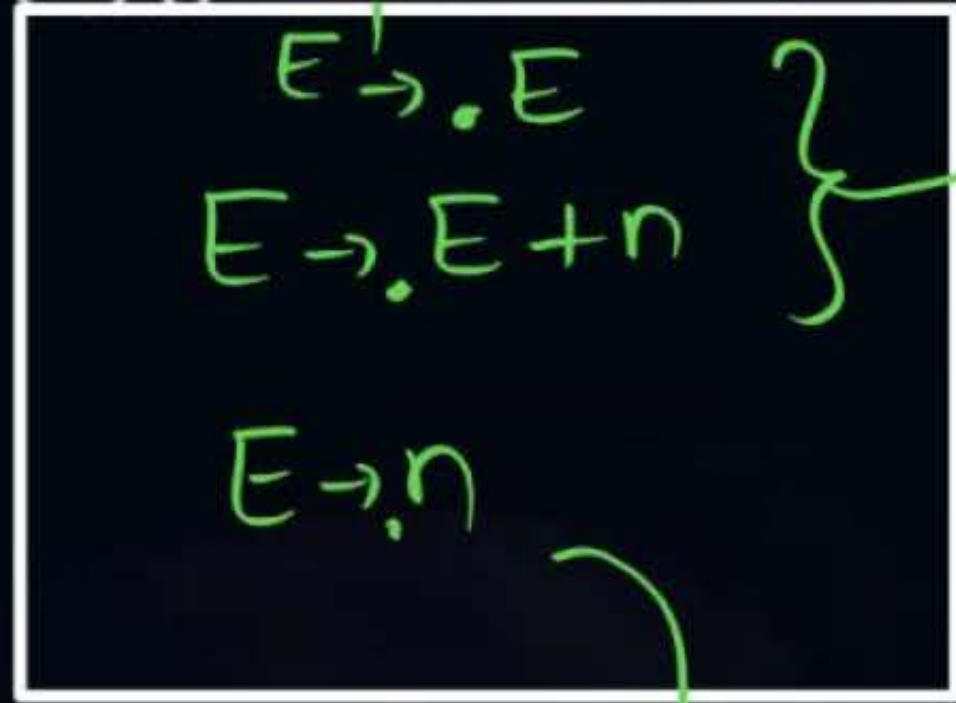
$B \rightarrow \cdot \epsilon \cdot (R)$

R/R Conflict

not LR(0) grammar

Σ_0

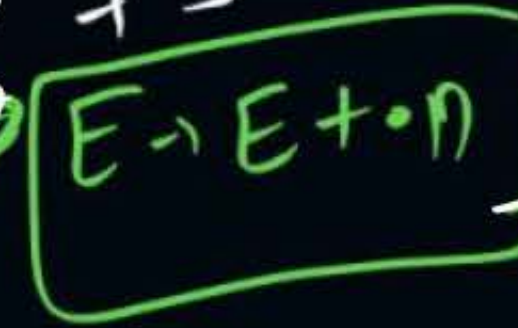
#Q. LR(0) grammar or not?



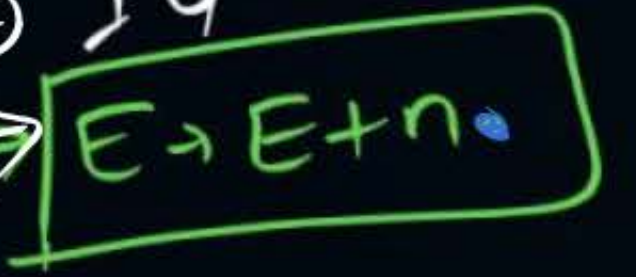
I_0



① I_3



② I_4



	+	n	\$
I_4	r_1	r_1	r_1
I_2	r_2	r_2	r_2

3 shift actions

GOTO = 1

Reduce actions = 6

- ① $E \rightarrow E + n$
- ② $E \rightarrow n$

is LR(0) grammar

#Q. LR(0) grammar or not?

Home Work

$$\left\{ \begin{array}{l} E \rightarrow E + T \\ E \rightarrow T \\ T \rightarrow i \end{array} \right\}$$



Topic : LR(0) Grammar Detection

- Any CFG of LR(0) parsing table contains
 - ① shift /reduce conflict (or)
 - ② reduce/reduce conflictthen give grammar is not LR(0) grammar

- S/R Conflict: state

In ~~terminal~~ shift

$$A \rightarrow \alpha \cdot a B \text{ (shift)}$$

$$B \rightarrow \gamma \cdot \text{ (Reduce)}$$

$A \rightarrow \alpha \cdot \underline{B} \times$

-
- R/R Conflict: state

In

$$A \rightarrow \alpha \cdot$$

$$B \rightarrow \gamma \cdot$$

Regular Grammar

CFG

Ambiguous Grammar

LR(0) grammar



2 mins Summary



Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU